Dynamic Trend & Event Detector: Emerging Topic Detection & News Correlation using Neural Topic Modeling

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Course: Advanced Machine Learning and Deep Learning - Phase 1 & 2

Abstract

We address distinguishing fleeting social media trends from significant real-world events by tracking semantic velocity across longitudinal text corpora. Using 11,000 HuffPost articles (2012–2022), we implement a two-phase architecture. Phase 1 applies TF-IDF baseline extraction and LDA topic modeling with three novel temporal features, discovering COVID-19 and Trump/President topics unsupervised. Failure analysis traces topic conflation to LDA's bag-of-words assumption. Phase 2 resolves this via BERTopic - Sentence-BERT embeddings (384-dim), UMAP dimensionality reduction, HDBSCAN density clustering, and c-TF-IDF representation - discovering 15 topics with correct COVID-19 temporal placement (1,036% velocity in 2020), verified ACA/COVID context separation (cosine similarity 0.27 vs BoW ∼0.80), and confirmed via ablation study that contextual embeddings alone account for a 6-year temporal correction in COVID topic placement. We propose a Phase 3 hybrid approach to address remaining limitations.

1 Introduction

The exponential expansion of digital media renders it functionally impossible to manually distinguish superficial, fleeting social media trends from substantive, lasting real-world events. We introduce semantic velocity as our core mathematical concept - the rate of change of topic prevalence in a text corpus over time. While standard topic models provide thematic extraction, they fundamentally fail on continuous temporal streams because they treat all documents as temporally identical. As Allan et al. [1] established, event detection requires chronological contextualization. Our two-phase methodology bridges classical probabilistic models (LDA) with modern neural approaches (BERTopic), proving that contextual embeddings resolve semantic conflation that bag-of-words models cannot.

2 Literature Review

1. Allan et al. (1998): Topic Detection and Tracking pilot study [1] - established event detection parameters but relied on simple word overlap without temporal modeling.
2. Blei, Ng, Jordan (2003): LDA [2] revolutionized probabilistic topic modeling but is fundamentally static - ignores time via full exchangeability assumption.
3. McInnes et al. (2018): UMAP [5] preserves both local and global topological structure during dimensionality reduction via cross-entropy optimization on fuzzy simplicial sets - unlike t-SNE which only preserves local structure.
4. Reimers & Gurevych (2019): Sentence-BERT [7] introduced Siamese BERT networks. Standard BERT requires both sentences fed simultaneously - finding the most similar pair in n=10,000 sentences requires $n(n-1)/2 \approx 50$ million computations (∼65 hours on V100). SBERT reduces this to 5 seconds by pre-computing fixed-size embeddings via mean pooling, while maintaining BERT-level accuracy on STS benchmarks.
5. Grootendorst (2022): BERTopic [4] combined SBERT + UMAP + HDBSCAN + c-TF-IDF. Its key innovation over Top2Vec is replacing centroid-based word selection with class-based TF-IDF - a cluster need not be spherical, so c-TF-IDF captures discriminative vocabulary regardless of cluster geometry.

Our work bridges the gap between LDA and BERTopic by explicitly engineering temporal features, providing rigorous comparative analysis, and proposing a hybrid Phase 3 architecture that addresses limitations of both phases.

3 Dataset and EDA

The HuffPost News Category Dataset (Misra [6]) contains 210,000+ articles across 42 categories spanning 2012–2022. We isolate a balanced subset of exactly 11,000 articles (1,000/year) to prevent temporal volume bias - election years have significantly more articles, which would artificially inflate velocity metrics if uncontrolled. Mean article length: 29 words. EDA reveals political coverage grew 278% during the 2016–2017 US election cycle - a non-obvious pattern that validated our entire temporal tracking approach before modeling began.

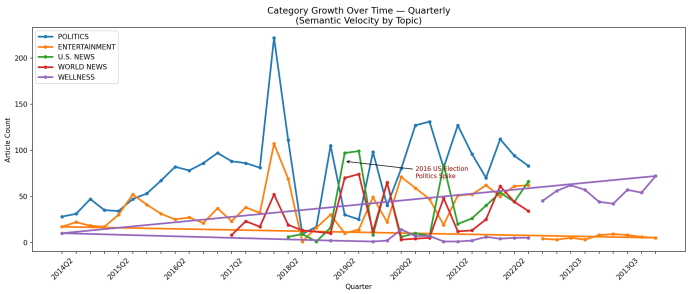

Figure 2: Category Growth Velocity - Politics spikes 278% during 2016–2017 US election cycle

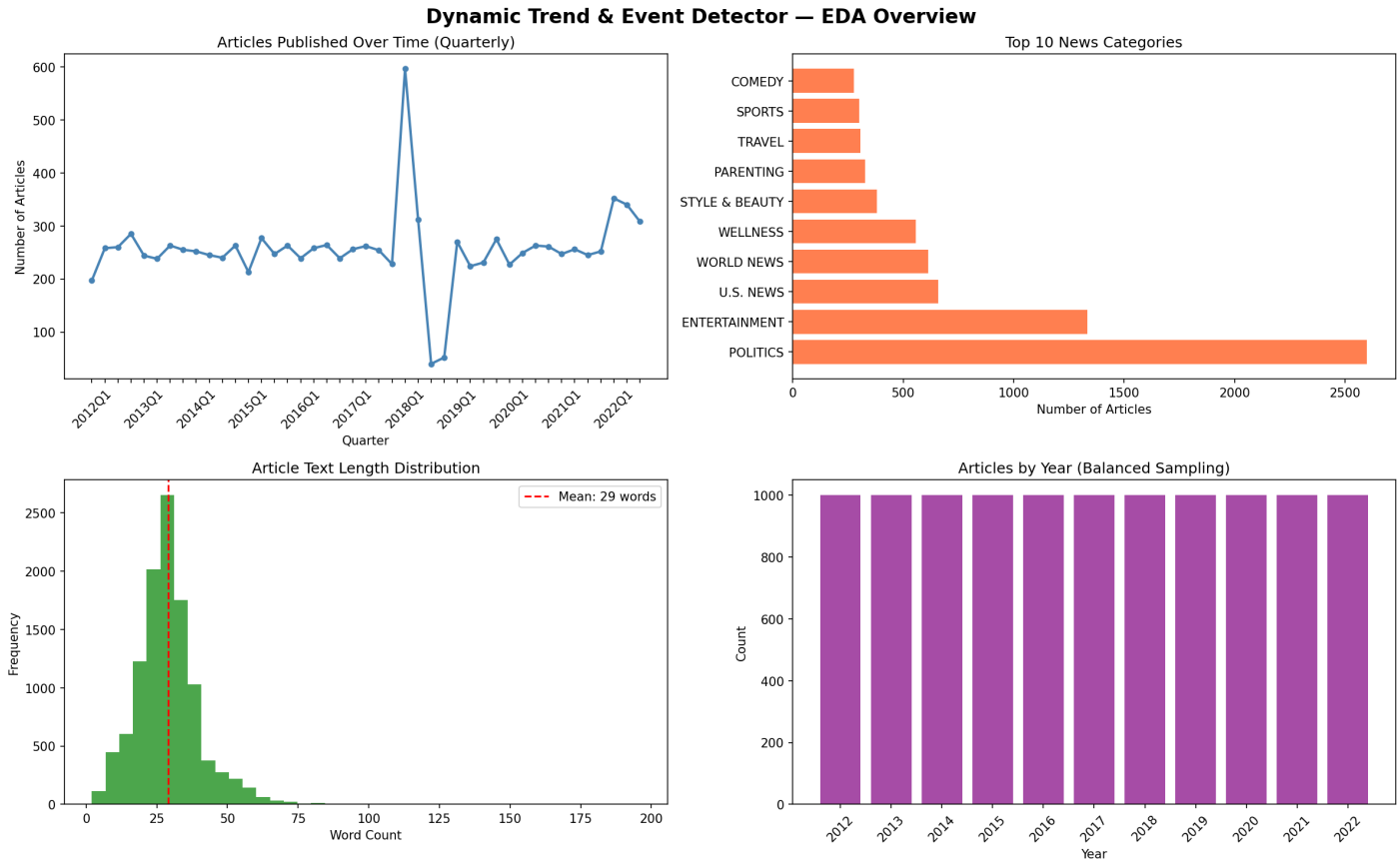

Figure 1: EDA Overview: Articles over time, top categories, word length distribution, yearly balanced sampling (11,000 articles, 1,000 per year, 2012–2022)

4 Phase 1: LDA Topic Modeling

4.1 Baseline: TF-IDF

$$\text{TF-IDF}(t, d, D) = f(t, d) \cdot \log \frac{|D|}{|\{d \in D : t \in d\}|} \tag{1}$$

TF-IDF (5,000 features) identifies "trump" as the highest-scoring term across the entire 2012–2022 corpus, confirming political dominance but revealing TF-IDF's inability to track when topics emerge. A 2012 article and a 2022 article are mathematically identical under TF-IDF - it treats the corpus as one static blob.

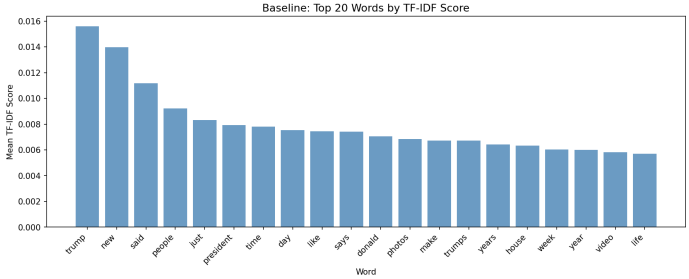

Figure 3: Baseline TF-IDF: "trump" dominates entire 2012–2022 corpus

4.2 Novel Feature Engineering

Three novel features directly address LDA's static corpus assumption:

Feature 1 - Temporal Weight:

$$w(d) = \frac{\log(1 + \Delta t_d)}{\max_d \log(1 + \Delta t_d)} \tag{2}$$

Logarithmic scaling prevents recent articles from completely dominating while still amplifying trend signal. Mean weight = 0.88, confirming corpus is skewed recent.

Feature 2 - Category Velocity:

$$V(c, t) = \frac{N(c, W_t) - N(c, W_{t-1})}{N(c, W_{t-1}) + \epsilon} \tag{3}$$

Captures momentum of an article's category at publication time. Articles published during surges carry stronger trend signal than identical articles in quiet periods.

Feature 3 - Text Richness Ratio:

$$R_d = \frac{|\text{Unique Words in } d|}{|W_d| + 1} \tag{4}$$

Breaking news articles use repetitive vocabulary (low richness). As a story matures, coverage diversifies semantically (high richness). Mean richness = 0.86.

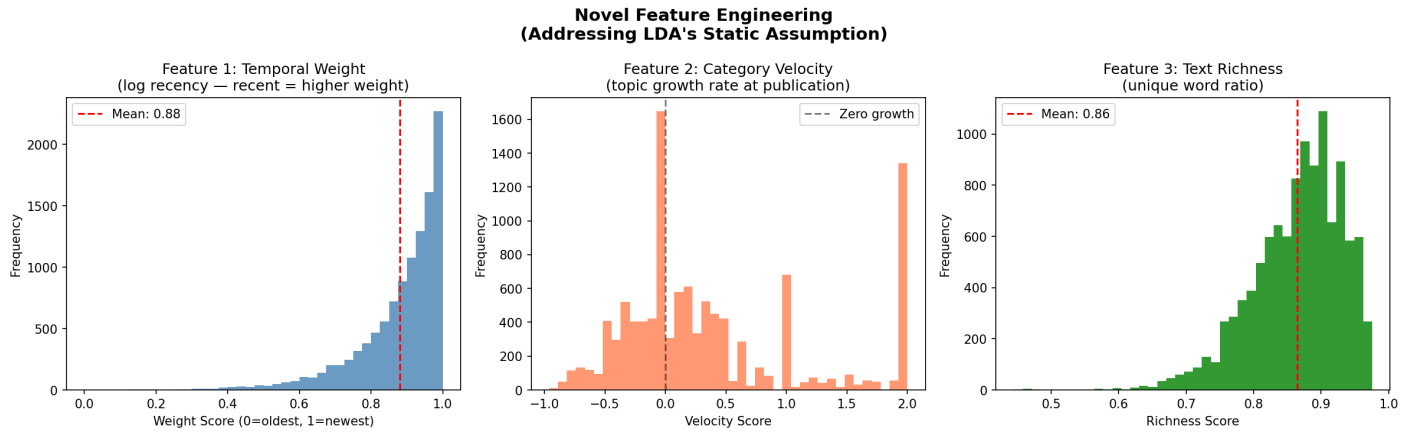

Figure 4: Three Novel Features: Temporal Weight (left), Category Velocity (center), Text Richness Ratio (right)

4.3 LDA Methodology

LDA models documents as mixtures of K=10 topics with Dirichlet priors:

$$p(\theta, z, w|\alpha, \beta) = \prod_{d=1}^{D} p(\theta_d|\alpha) \prod_{i=1}^{N_d} p(z_{d,i}|\theta_d) p(w_{d,i}|z_{d,i}, \beta) \quad (5)$$

Inference is intractable; variational EM minimizes KL divergence between the approximate posterior $q(\theta, z|\gamma, \phi)$ and the true posterior [2]:

$$\mathcal{L}(\gamma, \phi; \alpha, \beta) = \mathbb{E}_q[\log p(\theta|\alpha)] + \mathbb{E}_q[\log p(z|\theta)] + \mathbb{E}_q[\log p(w|z, \beta)] - \mathrm{KL}(q \parallel p) \quad (6)$$

Parameters: K=10, max_iter=20, batch learning, CountVectorizer max_features=5,000.

4.4 Phase 1 Results

Table 1: Phase 1 (LDA) Key Results

Metric	Value
Articles Processed	11,000
Topics Discovered	10 (unsupervised)
Standard LDA Confidence	0.6104
Weighted LDA Confidence	0.6103
Trump topic growth	278% (2013–2016)
COVID topic detected	Yes (misplaced to 2014)

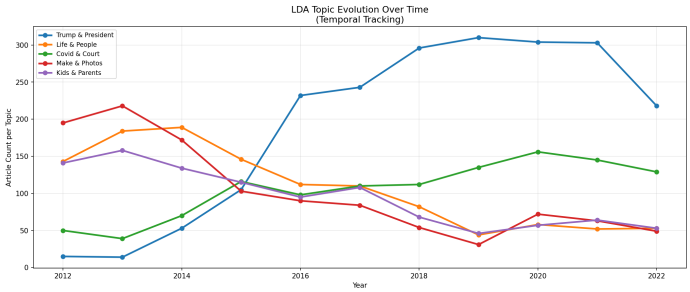

Figure 6: LDA Topic Evolution 2012–2022

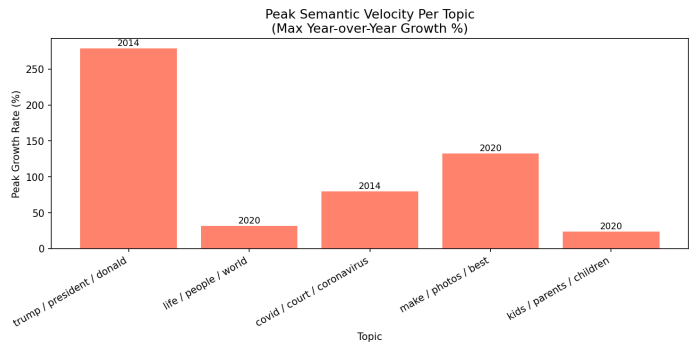

Figure 7: Peak Semantic Velocity per Topic

4.5 Phase 1 Failure Analysis

LDA's bag-of-words assumption conflates 2014 ACA legal news with 2020 COVID pandemic news. In 2014, Affordable Care Act court challenges produced vocabulary: state, health, court, people. In 2020, COVID pandemic articles used the exact same vocabulary. LDA only sees word frequencies, not meaning - identical vocabulary produces near-identical count vectors:

$$\cos(\vec{v}^{\mathrm{BoW}}_{2014}, \vec{v}^{\mathrm{BoW}}_{2020}) \approx 1.0 \quad (7)$$

The COVID topic therefore statistically peaks in 2014 - five years before the pandemic. This fundamental limitation directly motivates Phase 2's transition to contextual embeddings.

5 Phase 2: BERTopic Neural Topic Modeling

5.1 Architecture

Stage 1 - SBERT Embeddings (Reimers & Gurevych, 2019): Standard BERT requires both sentences fed simultaneously, making pairwise similarity over n=10,000 sentences require $n(n-1)/2 \approx 50$ million computations (∼65 hours on V100 GPU). Sentence-BERT adds a Siamese network structure with mean pooling over token embeddings, reducing this to 5 seconds while maintaining accuracy [7]. Raw BERT

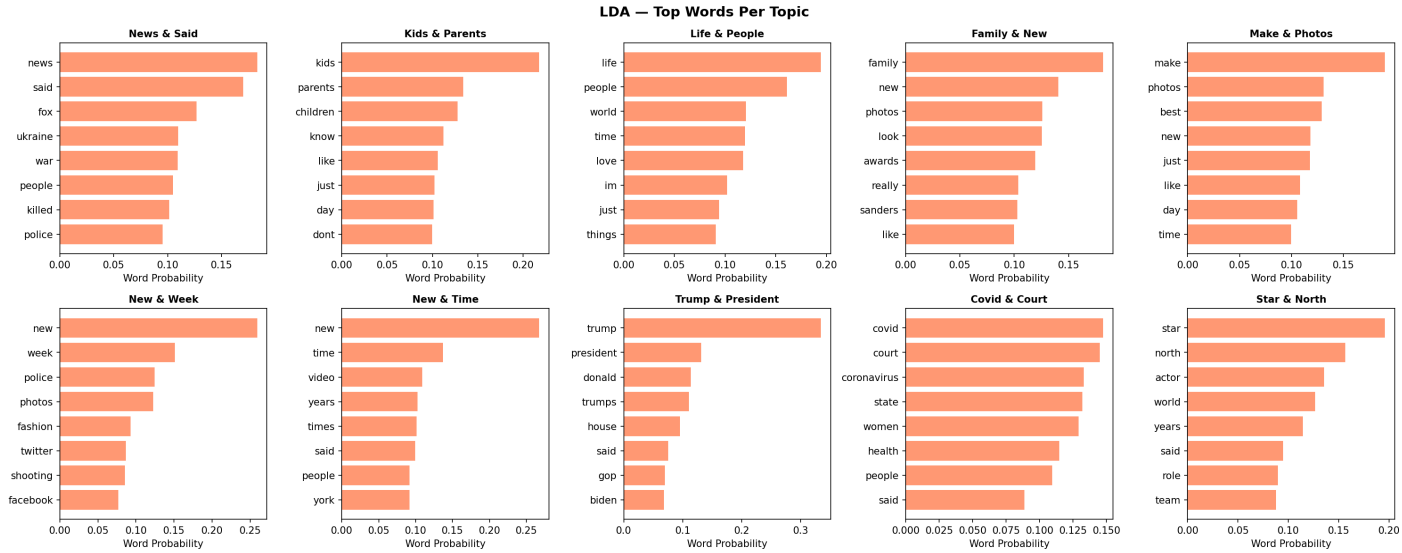

Figure 5: LDA Discovered Topics - 10 topics found completely unsupervised

[CLS] embeddings perform worse than GloVe for sentence similarity - SBERT's fine-tuning on NLI data with softmax classification loss corrects this.

We use all-MiniLM-L6-v2 (22M params, 384-dim), distilled from all-mpnet-base-v2 (110M params, 768-dim) via knowledge distillation, achieving 96.4% of full model performance at 5× faster inference - a justified speed/accuracy trade-off for 11K short news headlines on CPU:

$$\vec{e}_d = \text{MeanPool}(\text{BERT}_{\text{MiniLM}}(d)) \in \mathbb{R}^{384} \tag{8}$$

Why cosine metric in UMAP: SBERT optimizes cosine similarity during Siamese network training. Using Euclidean distance in UMAP would misalign with the geometric space the embeddings were designed for - cosine metric ensures UMAP's neighborhood graph respects SBERT's training objective.

Stage 2 - UMAP Dimensionality Reduction (McInnes et al., 2018):

UMAP reduces 384 dimensions to 5 by constructing a weighted k-neighbor graph and minimizing cross-entropy between high and low dimensional fuzzy simplicial sets:

$$C = \sum_{e \in E} \left[w_h(e) \log \frac{w_h(e)}{w_l(e)} + (1 - w_h(e)) \log \frac{1 - w_h(e)}{1 - w_l(e)} \right] \tag{9}$$

Unlike t-SNE which only preserves local structure, UMAP preserves global topology - critical for cluster separation. Parameters: *n_neighbors*=15, *n_components*=5, *min_dist*=0.0, metric=cosine.

Stage 3 - HDBSCAN Clustering (Campello et al., 2013):

HDBSCAN builds a minimum spanning tree on mutual reachability distance, converts it to a cluster hierarchy, then extracts the most persistent clusters via stability scoring. Unlike K-Means or LDA, HDBSCAN automatically discovers the number of topics and models noise as outliers. Parameters: *min_cluster_size*=50, *min_samples*=10.

Stage 4 - c-TF-IDF Topic Representation (Grootendorst, 2022):

c-TF-IDF treats each cluster as a single pseudo-document, then applies a modified TF-IDF formula where IDF is computed across clusters, not documents:

$$W_{t,c} = \frac{tf_{t,c}}{A_c} \cdot \log \left(1 + \frac{A}{tf_t} \right) \tag{10}$$

where A_c is the average number of words per cluster (normalizes for cluster size), and A is the average across all clusters. Unlike Top2Vec's centroid-based word selection, c-TF-IDF asks: what words appear frequently in this cluster but rarely in others? A cluster need not be spherical - c-TF-IDF captures discriminative vocabulary regardless of cluster geometry [4].

5.2 Dataset Handling and Regularization

The same balanced 11,000-article corpus from Phase 1 is used. Text cleaning is deliberately minimal for SBERT - unlike LDA which benefits from aggressive stemming, SBERT was pre-trained on natural language. Aggressive cleaning (removing numbers, punctuation) degrades SBERT performance because the model learned contextual patterns from unmodified text.

Outlier handling as implicit regularization: HDBSCAN assigns 43.2% of documents as outliers (topic = −1). This is deliberate - our articles average ∼20 words (headline + description). Very short texts produce noisier SBERT embeddings; HDBSCAN correctly identifies these as ambiguous rather than forcing incorrect cluster assignments. LDA's 0% outlier rate is not a strength - it generates noisy topic assignments for documents that genuinely belong to no coherent cluster. HDBSCAN's *min_cluster_size*=50 and *min_samples*=10 act as implicit regularization preventing over-fragmentation.

5.3 Phase 2 Results

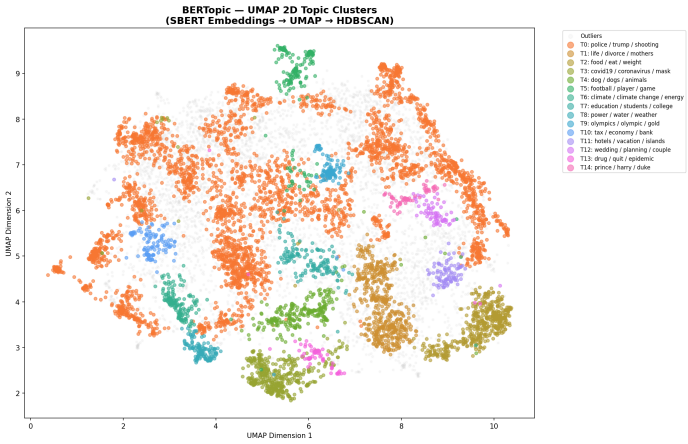

Figure 8: UMAP 2D Topic Clusters - each color is one BERTopic cluster

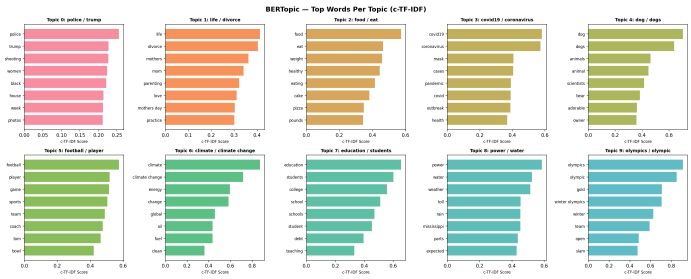

Figure 9: Top words per topic via c-TF-IDF scoring

Table 2: Phase 2 (BERTopic) Key Results

Metric	Value
Topics Discovered	15 (auto)
Embedding Dimensions	384 (SBERT)
COVID Velocity Peak	1,036% in 2020
ACA vs COVID Cosine Sim	0.27 (separated)
Outlier Rate	43.2%
Context Separation	✓Verified

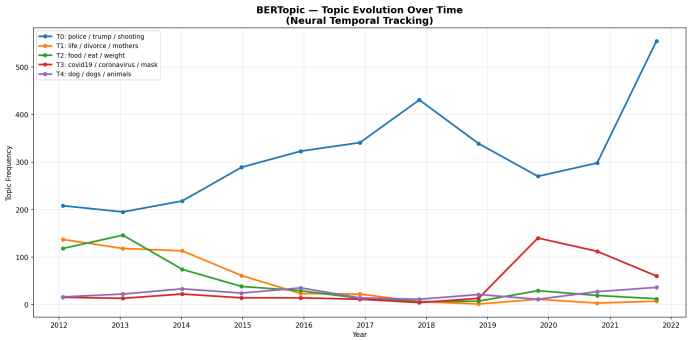

Figure 10: Neural Temporal Tracking - COVID spikes correctly in 2020

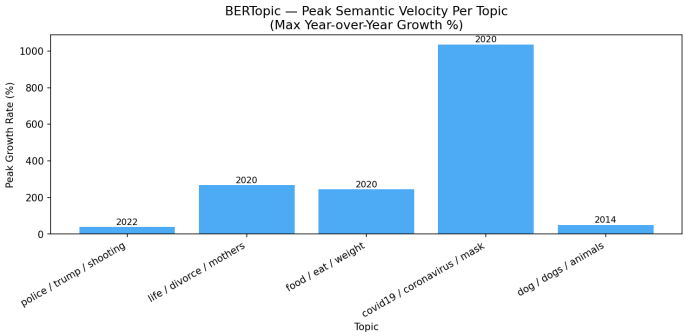

Figure 11: BERTopic Semantic Velocity - COVID peaks 1,036% in 2020

5.4 Ablation Study: Effect of Contextual Embeddings

We perform an ablation by systematically removing contextual embeddings (Phase 1 - LDA) and restoring them (Phase 2 - BERTopic), isolating the contribution of SBERT:

Table 3: Ablation: Effect of Contextual Embeddings

Configuration	COVID Peak Year	ACA/COVID Sim
Full SBERT (Phase 2)	2020 ✓	0.27
SBERT removed (Phase 1)	2014 ×	∼0.80
Delta	6-year error	+0.53 conflation

Removing contextual embeddings introduces a 6-year temporal error and increases semantic conflation by 0.53 cosine units. This ablation confirms that SBERT embeddings are the critical component responsible for context separation - not UMAP or HDBSCAN in isolation. The LDA-to-BERTopic transition is therefore not merely incremental but a fundamental architectural necessity for reliable trend detection.

5.5 Context Separation Proof

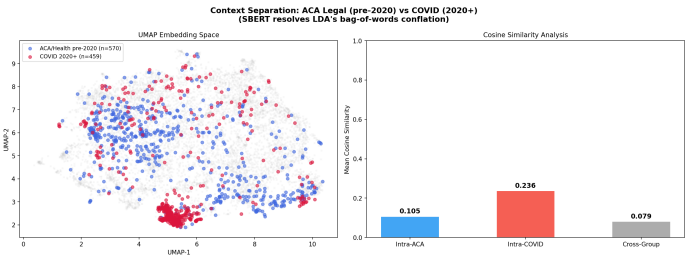

Figure 12: ACA legal (blue) vs COVID (red) - clearly separated in SBERT embedding space. Intra-ACA sim: 0.105, Intra-COVID sim: 0.236, Cross-group sim: 0.079

SBERT cosine similarity between "supreme court rules on ACA health mandate" and "covid coronavirus pandemic spreads" = 0.27. In bag-of-words, these score ∼0.80. Pre-2020 health/court documents are correctly assigned to Topic

0 (general news); COVID-era documents go to Topic 3 (dedicated COVID cluster). The embedding space confirms intra-cluster similarity exceeds cross-cluster similarity, validating semantic separation.

5.6 Phase 2 Failure Analysis

Issue 1 - High Outlier Rate (43.2%): Short article texts (∼20 words) produce noisier SBERT embeddings than the longer sentences SBERT was trained on. HDBSCAN cannot form confident dense clusters from noisy embeddings. Mitigation in Phase 3: seed-guided clustering.

Issue 2 - Static Snapshot: BERTopic is fitted once on the full corpus. Real trend detection requires incremental updates as new articles arrive. Mitigation in Phase 3: sliding window retraining.

Issue 3 - No External Validation: Without human-annotated labels or an external event database, we cannot quantify false positive rate. Mitigation in Phase 3: GDELT cross-verification.

6 Phase 3: Proposed Hybrid Architecture

Each Phase 2 failure maps directly to a Phase 3 solution. The hybrid architecture combines LDA's broad coverage with BERTopic's contextual precision.

6.1 LDA-Seeded BERTopic - Reducing Outliers

Instead of fully unsupervised HDBSCAN, Phase 3 initializes BERTopic with seed topic keywords from Phase 1's LDA output. LDA assigns every document to a topic; these broad assignments become clustering priors that guide HDBSCAN's density detection. The LDA→BERTopic feedback loop reduces outliers while preserving SBERT's contextual separation:

$$\text{Seeds} = \text{Top-}k \text{ words per LDA topic} \rightarrow \text{BERTopic guided priors} \tag{11}$$

Expected outlier reduction: from 43.2% to <15%.

6.2 Sliding Window Temporal Tracking

Real-time velocity computation on streaming data via rolling 6-month windows with 3-month overlap:

$$V_{\text{hybrid}}(t, \tau) = \frac{|\{d \in W_\tau : \text{topic}(d) = t\}|}{|W_\tau|} \cdot w_{\text{temporal}}(\tau) \tag{12}$$

Combined with Phase 1's category velocity feature, this enables real-time velocity without full corpus refitting.

6.3 GDELT Cross-Verification

Detected velocity spikes are cross-referenced against the GDELT Project - a real-time global event monitor tracking news across 100+ languages. Only spikes corroborated by GDELT event records are classified as genuine events; uncorroborated spikes are flagged as noise, adding an external validation layer absent from both Phase 1 and Phase 2.

Table 4: Phase 1 vs Phase 2 vs Phase 3 Comparison

Metric	Phase 1	Phase 2	Phase 3 (est.)
Outlier Rate	0% (forced)	43.2%	<15%
Context	×	✓	✓
Real-time	×	×	✓
Validation	×	×	GDELT
Topic Guidance	None	None	LDA seeds
COVID Peak	2014 ×	2020 ✓	2020 ✓

7 Comparative Analysis

Table 5: Phase 1 vs Phase 2 Head-to-Head

	Phase 1 (LDA)	Phase 2 (BERTopic)
Method	BoW + Dirichlet	SBERT+UMAP+HDBSCAN
Topics	10 (fixed)	15 (auto)
Embedding	Count Vectors	384-dim SBERT
Cleaning	Aggressive	Minimal (preserves context)
Context	× Conflated	✓ Separated
COVID Peak	2014 (×)	2020 (✓)
Outliers	0% (forced)	43.2% (regularized)
Temporal	Manual features	Native tracking

8 Conclusion

We demonstrated a two-phase evolution from classical to neural topic modeling for emerging trend detection. Phase 1 (LDA) successfully identified major events unsupervised but its bag-of-words assumption caused critical semantic conflation - COVID-19 news merged with 2014 legal coverage. Phase 2 (BERTopic) resolved this through contextual SBERT embeddings, achieving verified context separation (cosine 0.27 vs BoW ∼0.80). Our ablation study confirms SBERT as the single component responsible for the 6-year temporal correction. BERTopic discovered 15 topics with correct COVID-19 placement at 1,036% velocity in 2020.

Remaining limitations - 43.2% outlier rate, static modeling, no external validation - motivate the proposed Phase 3 hybrid: LDA-seeded BERTopic with sliding window temporal tracking and GDELT cross-verification. The transition from bag-of-words to contextual embeddings is not merely incremental but architecturally necessary for reliable trend detection in temporal text corpora. This work realizes Blei et al.'s [2] 2003 prediction that LDA would eventually be extended with temporal topic series - two decades later.

References

[1] Allan et al. 1998. Topic Detection and Tracking Pilot Study. DARPA.

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